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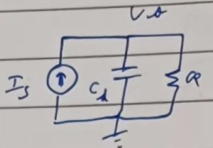
4.1:

Date _____
Page _____

Assignment - 4

3.3.2c

A1. a)



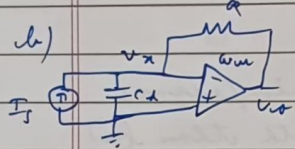
$$v_o = I_s \left(R \parallel \frac{1}{sC_d} \right) = I_s \frac{R}{1 + sRC_d}$$

$$\therefore BW = \omega_{-3dB} = \frac{1}{RC_d}$$

$$Gain = R \text{ at } s=0$$

$$\therefore Gain \times BW = \frac{1}{C_d}$$

b)



$$v_o = -v_x \times \frac{\omega_m}{s} \Rightarrow v_x = -v_o \times \frac{s}{\omega_m}$$

$$I_s = v_x \times sC_d + \frac{v_x - v_o}{R}$$

$$\therefore \frac{v_o}{I_s} = \frac{-\omega_m / C_d}{s^2 + \frac{s}{RC_d} + \frac{\omega_m}{RC_d}} = H(s)$$

$$s^2 + \frac{s}{RC_d} + \frac{\omega_m}{RC_d} = s^2 + 2\zeta\omega_n s + \omega_n^2; \quad \zeta = \frac{1}{2\zeta}$$

$$\omega_n = \sqrt{\frac{\omega_m}{RC_d}} \Rightarrow \zeta = \frac{1}{2\sqrt{\omega_m RC_d}}$$

$$\zeta = \frac{1}{\sqrt{2}} \Rightarrow \omega_m = \frac{1}{2RC_d}$$

$$\omega_{-3dB} \Rightarrow |H(s)|^2 = R^2/2 = \left| \frac{R}{1 - \frac{\omega^2 RC_d}{\omega_m} + j\omega/\omega_m} \right|^2$$

$$\therefore \omega^4 \left(\frac{R^2 C_d^2}{\omega_m^2} \right) + \omega^2 \left(\frac{1}{\omega_m^2} - \frac{2RC_d}{\omega_m} \right) - 1 = 0$$

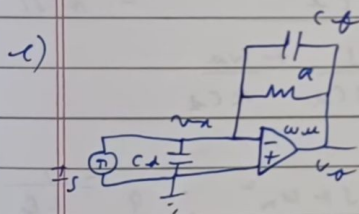
$$\omega^2 = \frac{2\alpha r_d \omega_m - 1 \pm \sqrt{8\alpha^2 C_d^2 \omega_m^2 - 4\alpha r_d \omega_m + 1}}{2\alpha^2 C_d^2}$$

For $\xi = \frac{1}{\sqrt{2}}$ ($\omega_m \alpha r_d = \frac{1}{2}$);

$$\omega^2 = \frac{1}{2\alpha^2 C_d^2} \Rightarrow \omega_{-3dB} = \frac{1}{\sqrt{2}} \times \frac{1}{\alpha C_d}$$

Gain at $\omega = \frac{1}{\sqrt{2}} \times \frac{1}{C_d}$

- Advantage $\rightarrow$ Better output impedance
- Disadvantage $\rightarrow$ Lesser bandwidth than fig



$$v_o = -v_x \times \omega_m / s$$

$$I_s = v_x s C_d + (v_x - v_o) \left(\frac{1}{\alpha} + s C_f \right)$$

$$\therefore v_o = -\omega_m / (C_d + C_f)$$

$$I_s = \frac{s^2 + s(1 + \omega_m \alpha C_f)}{\alpha (C_d + C_f)} + \frac{\omega_m}{\alpha (C_d + C_f)}$$

$$\therefore \omega_n = \sqrt{\frac{\omega_m}{\alpha (C_d + C_f)}}, \quad \xi = \frac{1 + \omega_m \alpha C_f}{2 \sqrt{\omega_m \alpha (C_d + C_f)}}$$

$$\xi = \frac{1}{\sqrt{2}} \Rightarrow C_f = \frac{\sqrt{2 \omega_m \alpha C_d} - 1}{\omega_m \alpha}$$

$$\omega_{-3dB} :- |H(s)|^2 = \alpha^2 / 2 = \left| \frac{\omega_m \alpha}{\omega_m - \omega^2 \alpha (C_d + C_f) + j \omega (1 + \omega_m \alpha C_f)} \right|^2$$

classmate

Date _____

Page _____

$$\therefore \omega^4 \alpha^2 (r_f + r_d)^2 + \omega^2 [(1 + \omega_m \alpha r_f)^2 - 2\alpha \omega_m (r_f + r_d)] - \omega_m^2 = 0$$

$$\therefore \omega^2 = \frac{-[(1 + \omega_m \alpha r_f)^2 - 2\alpha \omega_m (r_f + r_d)] \pm \sqrt{[(1 + \omega_m \alpha r_f)^2 - 2\alpha \omega_m (r_f + r_d)]^2 + 4\omega_m^2 \alpha^2 (r_f + r_d)^2}}{2\alpha^2 (r_f + r_d)^2}$$

For $\xi = \frac{1}{\sqrt{2}}$:- $\omega = \frac{\omega_m}{\alpha (r_f + r_d)}$

$$\omega = \frac{\omega_m}{\sqrt{\omega_m \alpha r_d + \sqrt{2\omega_m \alpha r_d - 1}}}$$

Gain $\times$ BW = $\frac{\omega_m \alpha}{\sqrt{\omega_m \alpha r_d + \sqrt{2\omega_m \alpha r_d - 1}}}$

- Advantage $\rightarrow$ The BW can be controlled by changing ω_m .
- Disadvantage $\rightarrow$ We cannot achieve an ^{allow} ~~arbitrary~~ ^{allow} ~~arbitrary~~ bandwidth lesser than $\omega_m < \frac{1}{2\alpha r_d}$.

Let's on plotting ~~the~~ ω , we see that there is also a minimum bandwidth that can be achieved.

4.2:

classmate

Date _____

Page _____

At 2. a)

$$g_m(v_i - v_o) + sC(v_i - v_o) = v_o s(C_o + C_L)$$

$$\therefore \frac{v_o}{v_i} = \frac{g_m + sC_i}{g_m + s(C_i + C_o + C_L)}$$

$\text{pole} = \frac{g_m}{C_i + C_o + C_L} = \omega_{ph}$

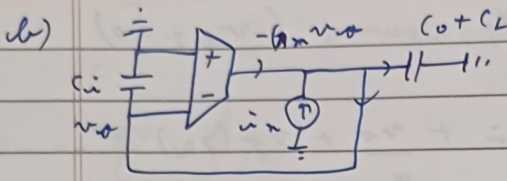
$$\therefore g_m = \omega_{ph}(C_i + C_o + C_L) = \omega_{ph} \left(\frac{2I_{DQ}}{V_{GS} - V_T} + C_L \right)$$

$$\therefore g_m = \frac{\omega_{ph} C_L \omega_T}{\omega_T - 2\omega_{ph}} \Rightarrow I_{DQ} = 0.5(V_{GS} - V_T) C_L \frac{\omega_T \omega_{ph}}{\omega_T - 2\omega_{ph}}$$

Inference :- Maximum pole frequency (bandwidth) is $\omega_T/2$ and to get a higher bandwidth, we need a higher bias current.

classmate
Date _____
Page _____

b)



$i_n = g_m v_o + v_o s (C_o + C_L + C_i)$
 $\therefore v_o = \frac{i_n}{g_m + s (C_o + C_L + C_i)}$
 $\therefore S_{v_o} = S_{i_n} = \frac{4kT g_m}{g_m^2 + [2\pi f (C_o + C_L + C_i)]^2} = \frac{4kT g_m}{g_m^2 + (2\pi f)^2 (C_o + C_L + C_i)^2}$

$\therefore \sigma^2 = \int_0^\infty S_{v_o} df = \frac{4kT g_m}{g_m^2} \int_0^\infty \frac{df}{1 + \left[\frac{2\pi f (C_o + C_L + C_i)}{g_m} \right]^2}$

$\sigma_{v_o}^2 = \frac{4kT}{g_m} \times \frac{1}{2\pi (C_o + C_L + C_i)} \int_0^\infty \frac{d \left[\frac{2\pi f (C_o + C_L + C_i)}{g_m} \right]}{1 + \left[\frac{2\pi f (C_o + C_L + C_i)}{g_m} \right]^2}$

$= \frac{4kT}{2\pi (C_o + C_L + C_i)} \times \frac{\pi}{2}$

$\sigma_{v_o}^2 = \frac{kT}{C_L} \times \frac{W_L}{W_T} = \frac{kT}{C_L} \left(\frac{W_T - 2W_L}{W_T} \right)$

$\therefore$ noise variance decreases as W_L increases

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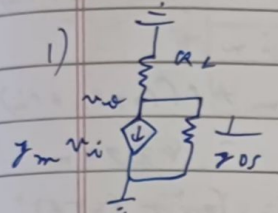
classmate

Date _____

Page _____

4.3. a) Bulb connected to source ($v_{ds} = 0$)

1)

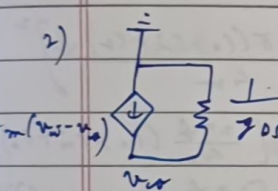


$$g_m v_i + \frac{v_o}{r_{DS}} = 0$$

$$\therefore \frac{v_o}{v_i} = -\frac{g_m R_L}{1 + g_{DS} R_L}$$

$\hookrightarrow R_{in} = \infty, \alpha_{out} = \frac{1}{g_{DS}}$

2)



$$g_m (v_i - v_o) = v_o \times g_{DS}$$

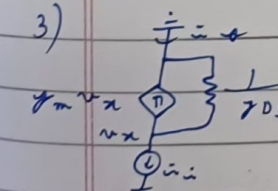
$$\therefore \frac{v_o}{v_i} = \frac{g_m}{g_m + g_{DS}}$$

$\hookrightarrow R_{in} = \infty$

$\hookrightarrow \therefore \alpha_{out} = \left(\frac{1}{g_m} \parallel \frac{1}{g_{DS}} \right)$

$\therefore \alpha_{out} = \frac{1}{g_m + g_{DS}}$

3)

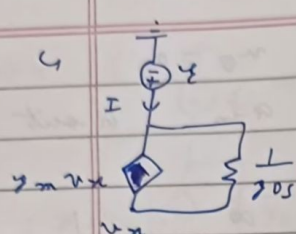


$$v_o = v_i$$

$$\therefore \frac{v_o}{v_i} = 1$$

$\hookrightarrow R_{in} = \frac{1}{g_m + g_{DS}}$

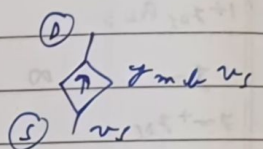
Date _____
 Page _____

1) 

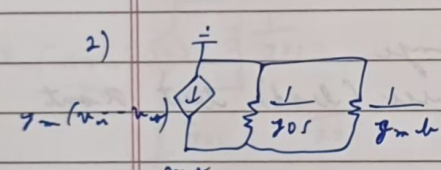
$$I + g_m v_x + (v_x - 0) g_{o5} = 0$$

$$g_m v_x + (v_x - 0) g_{o5} = 0$$

$$\therefore I = 0 \Rightarrow \alpha_{out} = \infty$$

2) $v_{BS} = v_B - v_S = -v_S \Rightarrow$ 

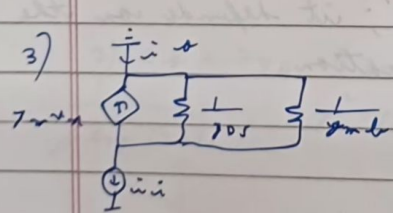
1) $v_S = 0$ always
 $\therefore v_{BS} = 0 \Rightarrow$ same circuit for gain, α_{in} , α_{out}

2) 

$$g_m(v_i - v_o) = v_o(g_{o5} + g_{mb})$$

$$\frac{v_o}{v_i} = \frac{g_m}{g_m + g_{o5} + g_{mb}}$$

$\therefore \alpha_{in} = \infty, \alpha_{out} = \frac{1}{g_m + g_{o5} + g_{mb}}$

3) 

$$\frac{i_o}{i_i} = 1 = \alpha_{in}$$

$\therefore \alpha_{in} = \frac{1}{g_m + g_{o5} + g_{mb}}, \alpha_{out} = \infty$

classmate						
Date _____						
Page _____						
1) $v_0 = v_s$			2) $v_0 = 0$			
G_{in}	R_{in}	α_{out}	G_{in}	R_{in}	α_{out}	
a)	$\frac{-g_m R_L}{1 + g_{DS} R_L}$	∞	$\frac{-g_m R_L}{1 + g_{DS} R_L}$	∞	$\frac{1}{g_{DS}}$	
b)	$\frac{g_m}{g_m + g_{DS}}$	∞	$\frac{g_m}{g_m + g_{DS} + g_{mB}}$	∞	$\frac{1}{g_m + g_{DS} + g_{mB}}$	
c)	$\frac{1}{g_m + g_{DS}}$	∞	$\frac{1}{g_m + g_{DS} + g_{mB}}$	∞	$\frac{1}{g_m + g_{DS} + g_{mB}}$	
- In circuit a), no change. - In circuit b), G_{in} ↓ (bad) but α_{out} also ↓ (good) - In circuit c), α_{in} ↓ (good)						
Hence, we cannot objectively say whether body effect is 'good' or 'bad'; it depends on the circuit and the application.						

4.4:

classmate

Date _____

Page _____

A4. a)

$$I + \frac{g_m I}{r_{DS}} = \left(\frac{V - I}{r_{DS}} \right) r_{DS}$$

$$\therefore z_{out} = \frac{V}{I} = \frac{2r_{DS} + g_m}{g_m^2 r_{DS}}$$

b)

$$I + g_m(A+1) \frac{I}{r_{DS}} = \left(\frac{V - I}{r_{DS}} \right) r_{DS}$$

$$\therefore z_{out} = \frac{V}{I} = \frac{2r_{DS} + g_m(A+1)}{g_m^2 r_{DS}}$$

1) $z_{out} = \frac{2r_{DS} + g_m \left(\frac{\omega_u}{s} + 1 \right)}{g_m^2 r_{DS}} = \frac{g_m \omega_u + s(g_m + 2r_{DS})}{s g_m^2 r_{DS}}$

2) $z_o = \frac{2r_{DS} + g_m \left(\frac{\omega_u}{s + \omega_1} + 1 \right)}{g_m^2 r_{DS}} = \frac{g_m \omega_u + \omega_1(g_m + 2r_{DS}) + s(g_m + 2r_{DS})}{g_m^2 r_{DS} (s + \omega_1)}$

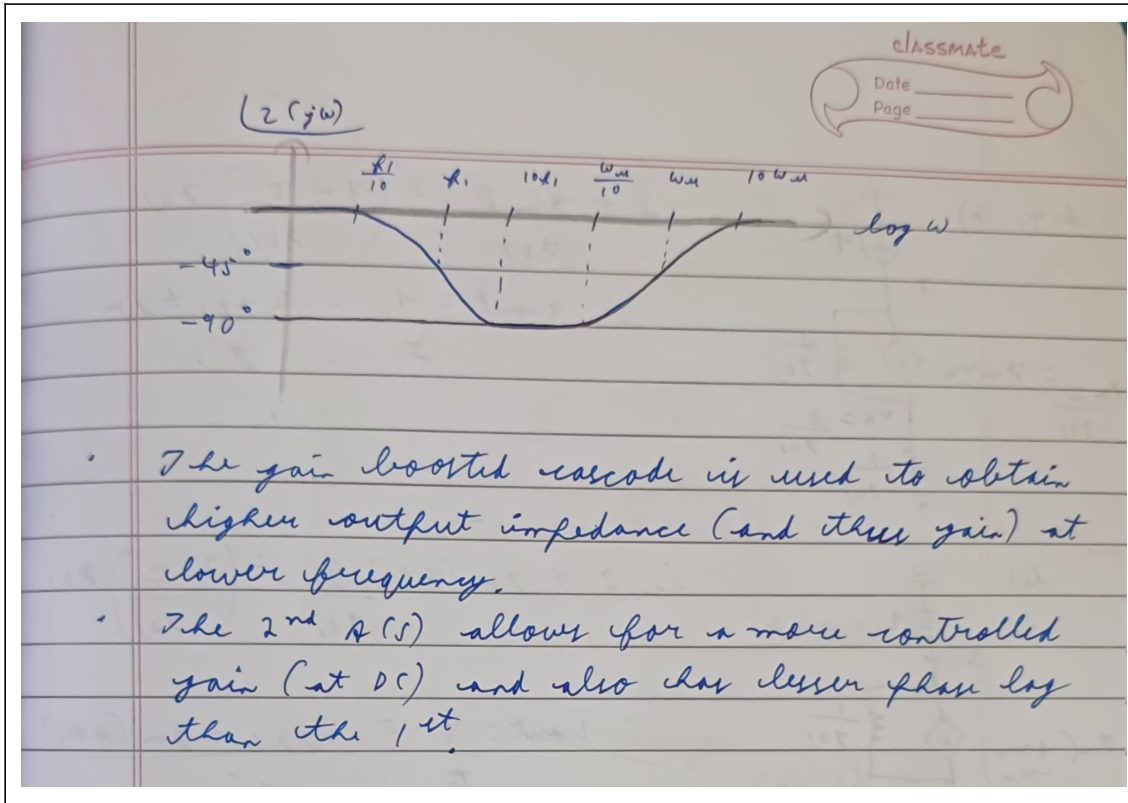
$\log |z+1|$

$\log \omega$

$\omega_u + \omega_1 = \omega_u + \frac{\omega_u}{\omega_0}$

$\approx \omega_u$

$\approx \omega_u$



4.5:

classmate
Date _____
Page _____

Ans.
$$L(s) = \frac{w_{u, \text{loop}}}{s} \left[\frac{1 + b_1 s + \dots + b_m s^m}{1 + a_1 s + \dots + a_n s^n} \right]$$

$$L(s) = \frac{w_{u, \text{loop}}}{s} + \frac{w_{u, \text{loop}}}{s} \left[\frac{(b_1 - a_1) s + \dots + (b_m - a_m) s^{m+1} - a_{m+1} s^{m+1} - \dots - a_n s^n}{1 + a_1 s + \dots + a_n s^n} \right]$$

$$L(s) = \frac{w_{u, \text{loop}}}{s} + \frac{w_{u, \text{loop}}}{s(1 + a_1 s + \dots + a_n s^n)} \left[(b_1 - a_1) + \dots + (b_m - a_m) s^{m+1} - a_{m+1} s^{m+1} - \dots - a_n s^n \right]$$

↓
ramp with slope $w_{u, \text{loop}}$

To find the steady state response of the 2nd term, we use final value theorem:-

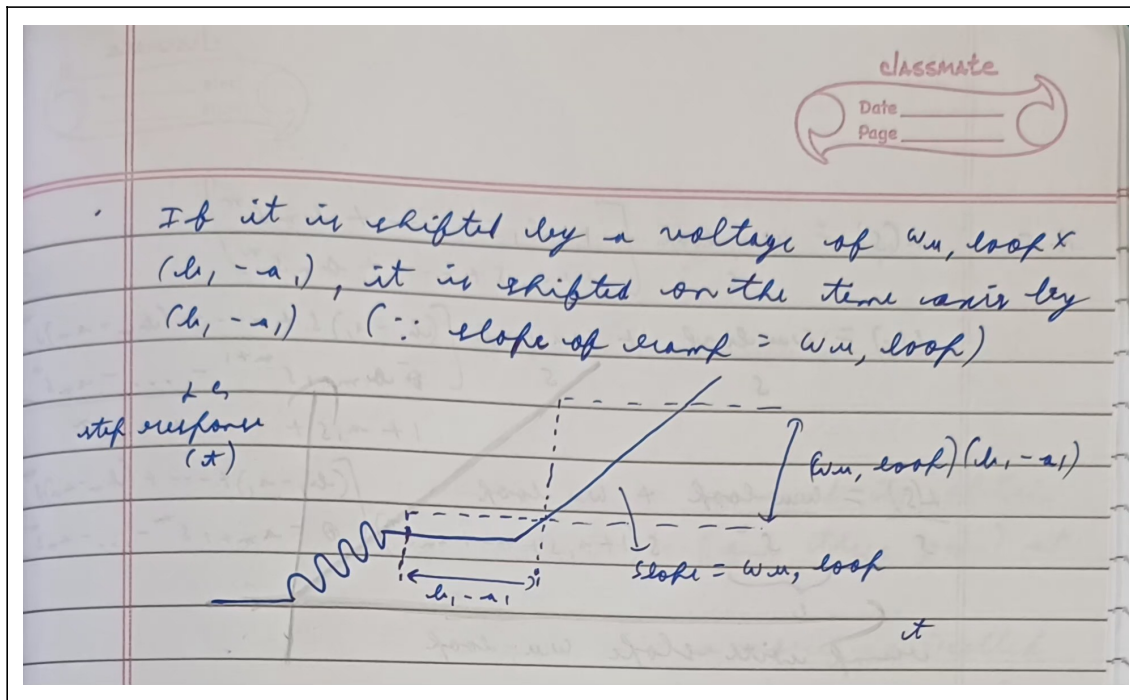
$$\text{Final value} = \lim_{s \rightarrow 0} \left(s \times \frac{L(s)}{s} \right) = \lim_{s \rightarrow 0} \left[\frac{w_{u, \text{loop}}}{1 + a_1 s + \dots + a_n s^n} \left[(b_1 - a_1) + \dots + (b_m - a_m) s^{m+1} - a_{m+1} s^{m+1} - \dots - a_n s^n \right] \right]$$

$$\text{Final value} = w_{u, \text{loop}} (b_1 - a_1)$$

Upon close observation, we see that $b_i = \sum \frac{1}{z_i}$ and $a_i = \sum \frac{1}{p_i}$ where z_i and p_i are the zeros and poles respectively.

$$\therefore 1 + b_1 s + \dots + b_m s^m = \left(1 + \frac{s}{z_1}\right) \left(1 + \frac{s}{z_2}\right) \dots \left(1 + \frac{s}{z_m}\right)$$

and $1 + a_1 s + \dots + a_n s^n = \left(1 + \frac{s}{p_1}\right) \left(1 + \frac{s}{p_2}\right) \dots \left(1 + \frac{s}{p_n}\right)$

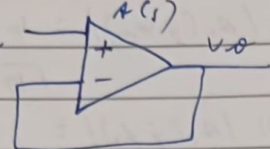


4.6:

classmate

Date _____

Page _____

As a) 

$$v_o = A(v_i - v_o)$$

$$\therefore \frac{v_o}{v_i} = \frac{A}{A+1}$$

} closed loop transfer function

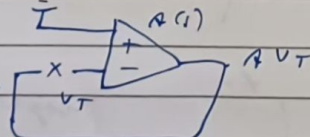
1)
$$\frac{v_o}{v_i} = H(s) = \frac{\omega_n}{\omega_n + s\left(1 + \frac{s}{\zeta\omega_n}\right)}$$

$$= \frac{\omega_n \zeta\omega_n}{s^2 + s\zeta\omega_n + \omega_n^2}$$

• $\omega_n = \sqrt{\omega_n \zeta\omega_n} \Rightarrow \zeta = \frac{\zeta\omega_n}{2\omega_n} = \frac{\zeta\omega_n}{2\omega_n}$

2)
$$H(s) = \frac{\omega_n(\zeta_1 + s)}{s^2 + s\omega_n + \omega_n\zeta_1}$$

• $\omega_n = \sqrt{\omega_n\zeta_1} \Rightarrow \zeta = \frac{\omega_n}{2\omega_n} = \frac{\omega_n}{2\omega_n}$

b) 

$$\therefore L(s) = A(s)$$

• critically damped $\Rightarrow \zeta = 1 \Rightarrow \omega_n = \frac{\zeta\omega_n}{\zeta} = \frac{\zeta\omega_n}{1}$

